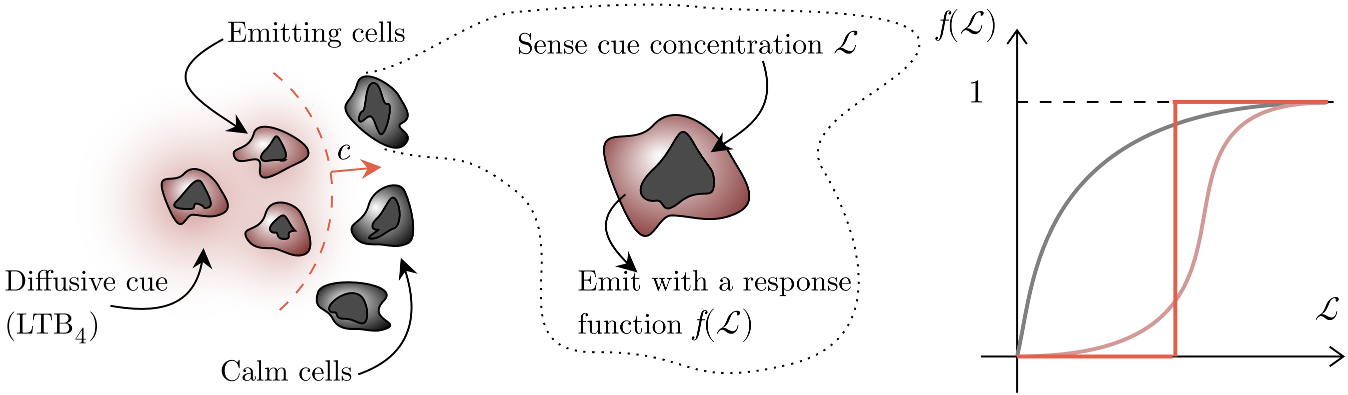
Diffusive relay model

Prior work: *Dieterle et al., 2020; Dieterle & Amir, 2021*



$$\partial_t \mathcal{L} = D\Delta\mathcal{L} + \alpha f(\mathcal{L}) \sum_i \delta(\mathbf{r} - \mathbf{r}_i) \quad \Leftrightarrow \quad \partial_t \mathcal{L} = D\Delta\mathcal{L} + \alpha\rho f(\mathcal{L}).$$

diffusivity of the signaling molecule

i^{th} neutrophil position

maximal emission rate per cell

average cell density

Diffusive relay model

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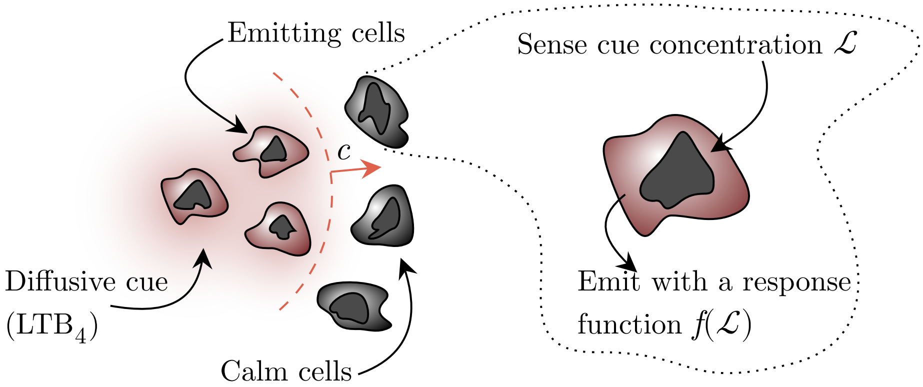
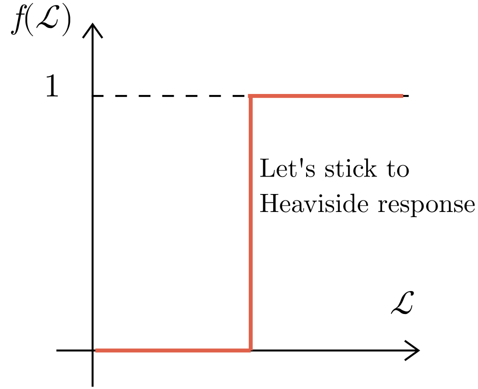




$$\partial_t \mathcal{L} = D\Delta\mathcal{L} + \alpha\rho\Theta(\mathcal{L} - \mathcal{L}_0) \quad \overset{\mathcal{L}(r,t)=f(r-ct)}{\Longrightarrow} \quad D\partial_r^2 f + c\partial_r f + \alpha\rho\Theta(f - f_0) = 0$$

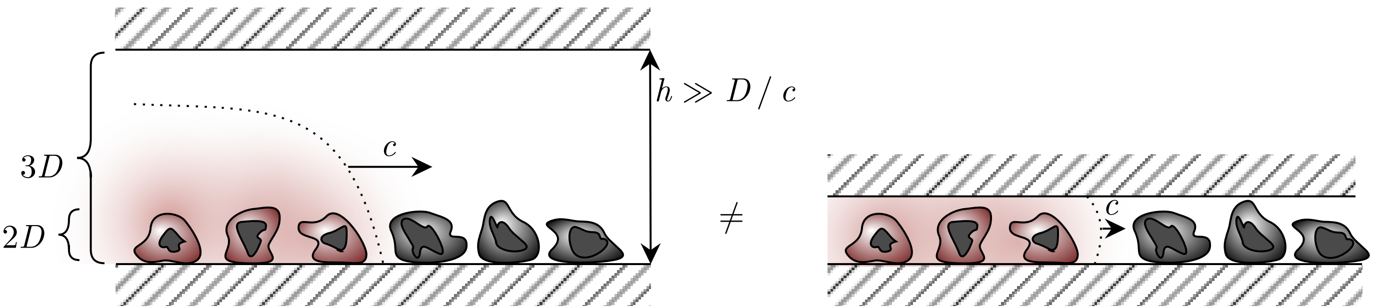
This has traveling waves, similar to Fisher-KPP $\partial_t n = \partial_x^2 n + n(1-n)$

Dimensionality effects

Prior work: *Dieterle et al., 2020; Dieterle & Amir, 2021*

$$\partial_t \mathcal{L} = D\Delta\mathcal{L} + \alpha\rho\Theta(\mathcal{L} - \mathcal{L}_0)$$

- '2D-3D' and '2D-2D' setups result in different wave speed c
- Curvature term in the Laplacian is negligible: '2D-2D' is similar to '3D-3D'



$$c = \frac{2\alpha\rho}{\pi\mathcal{L}_0}$$

$$c = \sqrt{\frac{\alpha\rho D}{h\mathcal{L}_0}}$$

Self-extinguishing waves

Prior work:

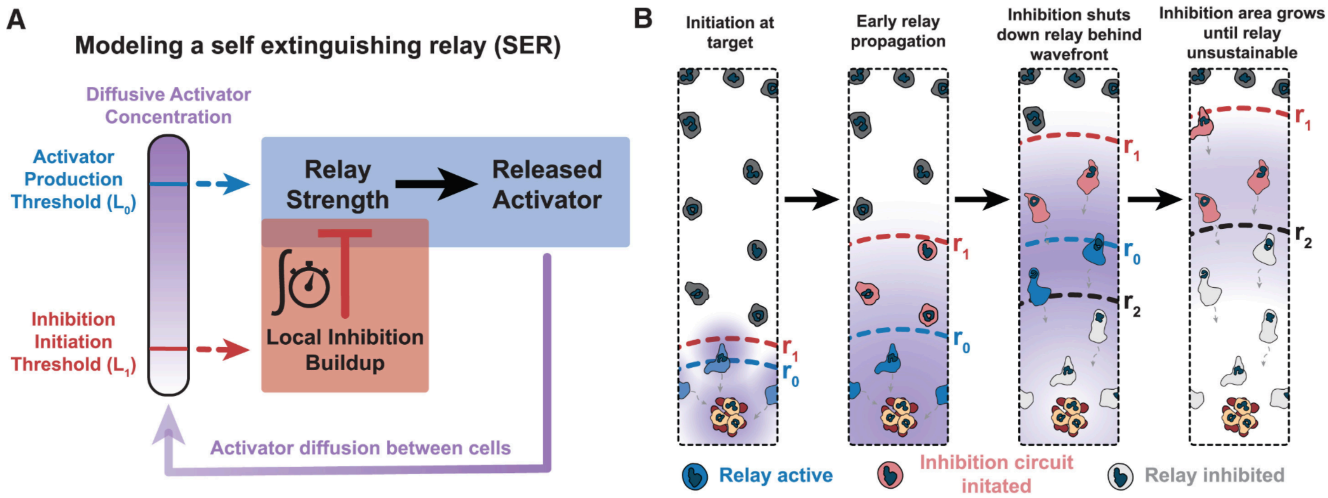
Strickland, Pan, ..., Ji,..., Amir, Weiner, 2024

our group

$$\partial_t \mathcal{L} = \underbrace{D\Delta\mathcal{L}}_{\text{diffusion}} + \underbrace{\alpha\rho(\mathbf{r})\Theta(1-\mathcal{R})\Theta(\mathcal{L}-1)}_{\text{activation with } \mathcal{L}\text{, inhibition with } \mathcal{R}} \quad \underbrace{-\Gamma_L\mathcal{L}}_{\text{active degradation}}$$

$$\partial_t \mathcal{R}_i = \underbrace{\beta\Theta\big(\mathcal{L}(\mathbf{r}_i) - \mathcal{L}_r\big)}_{\text{activation with } \mathcal{L}} \quad \underbrace{-\Gamma_R\mathcal{R}_i}_{\text{active degradation}}$$

index enumerates neutrophils



Self-extinguishing waves

Prior work:

Strickland, Pan, ..., Ji,..., Amir, Weiner, 2024

Deng's video here

Inhibitor model in 2D-2D: analytical approach

My work

Centrally-symmetric setup, neutrophils are uniformly distributed.

$$\boxed{\begin{aligned}\partial_t\mathcal{L} &= \Delta_r\mathcal{L} + \alpha\Theta(1-\mathcal{R})\Theta(\mathcal{L}-1) - \mathcal{L}\\ \partial_t\mathcal{R} &= \beta\Theta(\mathcal{L}-\mathcal{L}_r) - \gamma\mathcal{R}\end{aligned}}$$

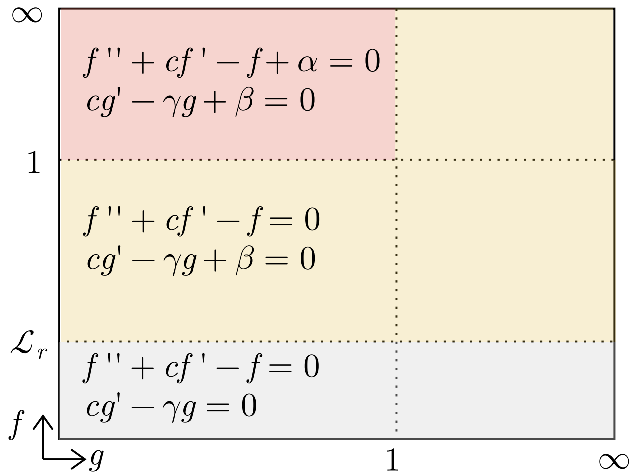
$$\downarrow \mathcal{L}(r,t) = f(r-ct),\ \ \mathcal{R}(r,t) = g(r-ct)$$

$$\boxed{\begin{aligned}-c\partial_r f &= \Delta_r f + \alpha\Theta(1-g)\Theta(f-1) - f\\ -c\partial_r g &= \beta\Theta(f-\mathcal{L}_r) - \gamma g\end{aligned}}$$

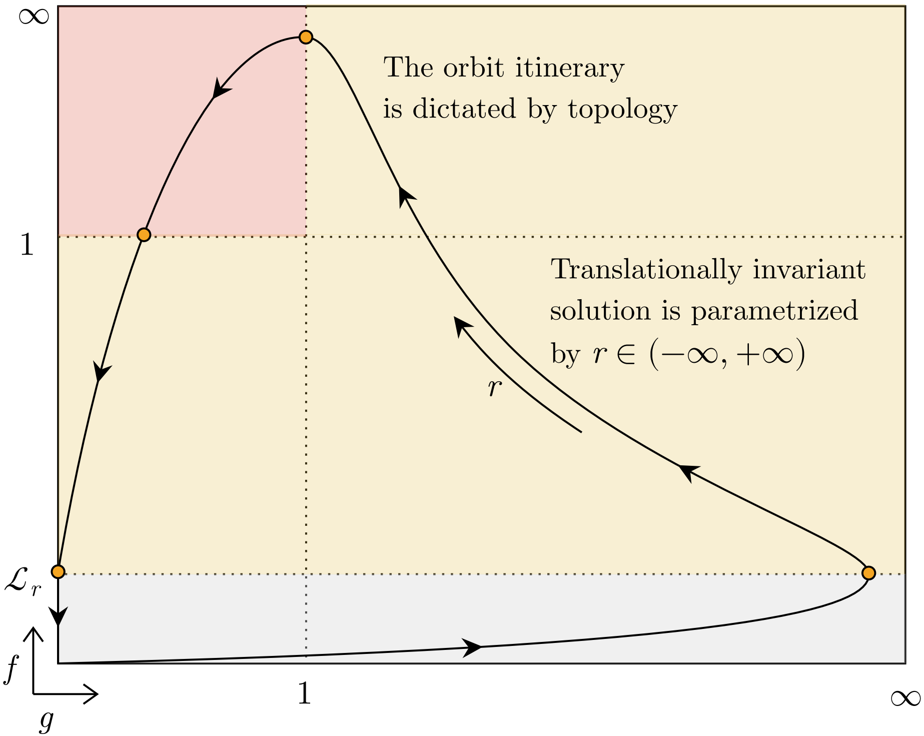
$\downarrow$ neglecting the curvature term in the radial Laplacian

$$\boxed{\begin{aligned}&f'' + cf' - f + \alpha\Theta(1-g)\Theta(f-1) = 0,\\ &cg' - \gamma g + \beta\Theta(f-\mathcal{L}_r) = 0.\end{aligned}}$$

$\leftarrow$ piecewise translational invariance



Inhibitor model in 2D-2D: analytical approach



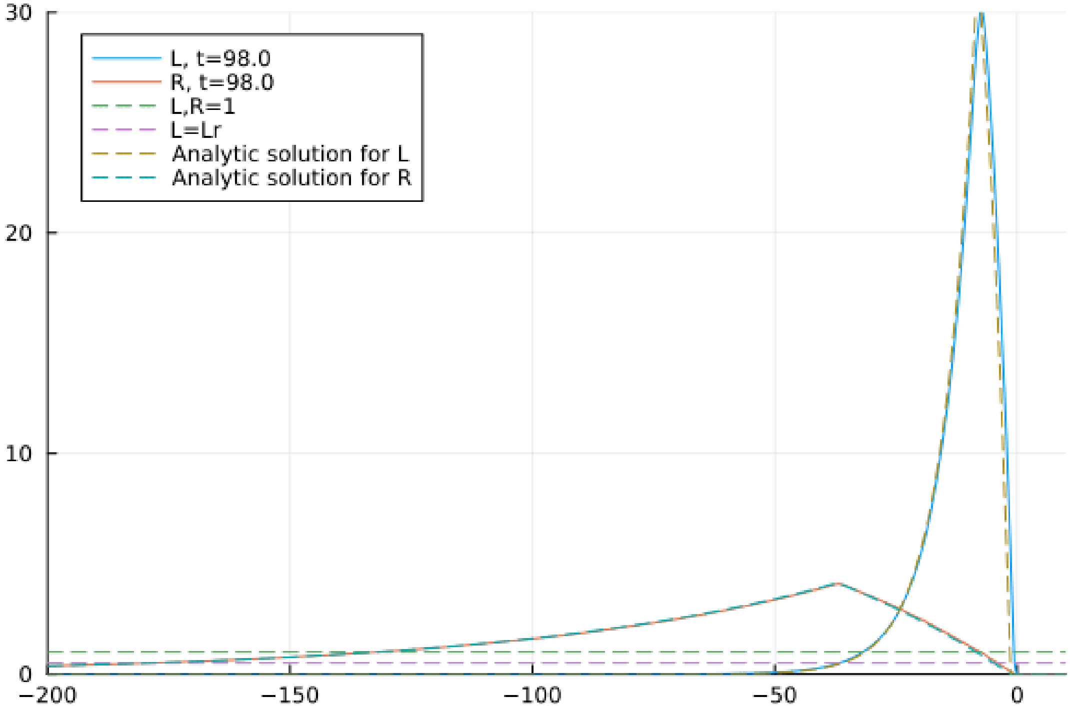
$$f'' + cf' - f + +\alpha\Theta(1-g)\Theta(f-1) = 0,$$

$$cg' - \gamma g + \beta\Theta(f - \mathcal{L}_r) = 0.$$

- The subtle point is determining the orbit shape, was done with monotonicity analysis
- Full solution is obtained by QM-like matching
- The matching fixes positions r of the threshold-crossing points and the wave speed c

Analytical solution vs simulations

My work



Adressing neutrophil motility

$$\partial_t \mathcal{L} = \underbrace{\Delta \mathcal{L}}_{\text{diffusion}} + \underbrace{\alpha \rho(\mathbf{r}) \Theta(1 - \mathcal{R}) \Theta(\mathcal{L} - 1)}_{\text{activation with } \mathcal{L}\text{, inhibition with } \mathcal{R}} \underbrace{- \Gamma_L \mathcal{L}}_{\text{active degradation}} ,$$

$$\partial_t \mathcal{R}_i = \underbrace{\beta \Theta\Big(\mathcal{L}(\mathbf{r}_i) - \mathcal{L}_r\Big)}_{\text{activation with } \mathcal{L}} \underbrace{- \Gamma_R \mathcal{R}_i}_{\text{active degradation}}$$

$\mathbf{r}_i$ – position of i^{th} neutrophil

$\boxed{\text{swarming gif here}}$

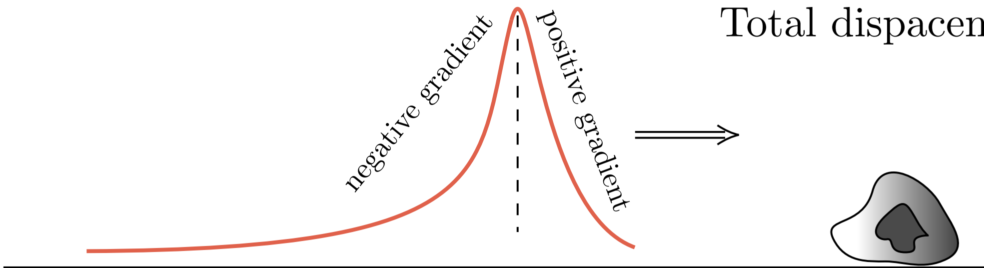
Chemotactic movement:

$$\mathbf{v}_i \equiv \frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} = ?$$

Seminal chemotaxis model: Keller-Segel

$$\frac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} = \mu \nabla \mathcal{L} + \xi \; - \; \text{velocity is propotional to the cue gradient}$$

Can't work with waves!



Total dispacement is *zero*

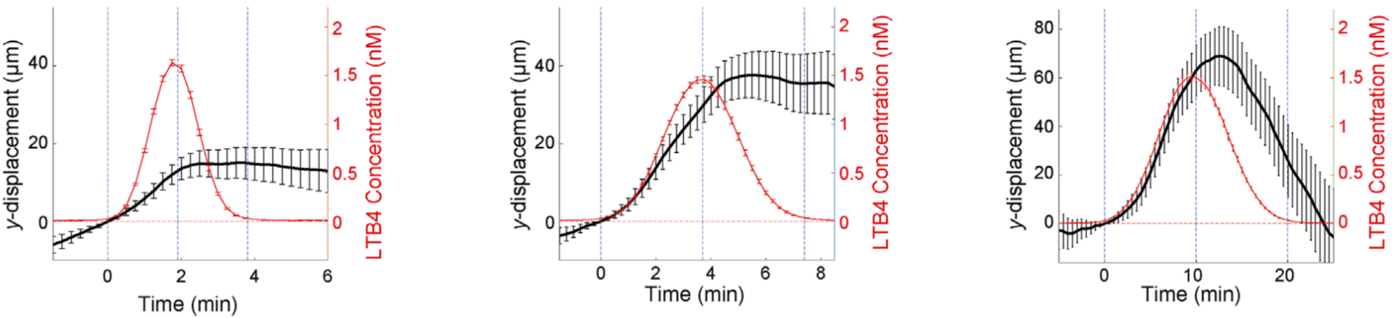
Experiments with HL-60 cells

Ishida et al., 2025

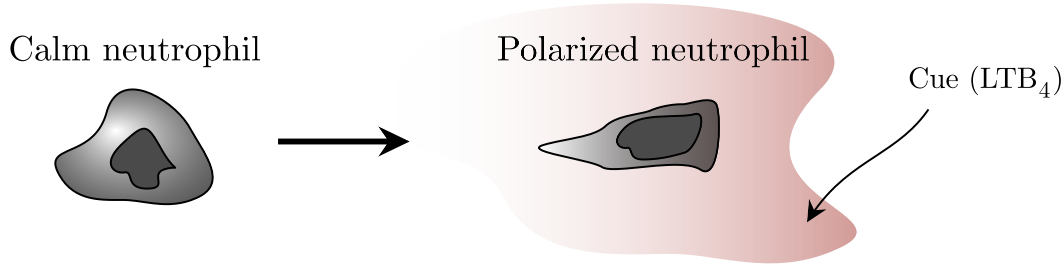
- Differentiated HL-60 cells are a standard model for human neutrophils (similar chemotaxis and polarity).

- Experiment: impose artificial fMLP/LTB_4 traveling waves with controlled speed using microfluidics.

- Cells chemotax only for intermediate wave speeds: too fast → no polarization, too slow → compensation.



Minimal model: polarization-induced chemotaxis



Introduce a vector polarization variable $\mathbf{p}$. Dynamics:

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \varkappa \nabla \mathcal{L} - \gamma_p \mathbf{p}, \qquad \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mu \boldsymbol{\sigma}(\mathbf{p}).$$

$\boldsymbol{\sigma}(\mathbf{p}) =$

- $\mathbf{p}$, "inertial" chemotaxis, maps to Newtone's law
- $\dfrac{p^n}{p_0^n + p^n} \dfrac{\mathbf{p}}{p}$, Hill-type response, has a velocity limit

Hill-type chemotaxis works

Both swarming and self-extinguishing behavior are reproduced

Hill-type gif here

CGD and clumping

- Chronic Granulomatous Disease (CGD) neutrophils form dense clumps instead of organized swarms.
- This clumping phenotype is reproducible in vitro by blocking the NADPH-oxidase inhibitor pathway.
- In the two-threshold model, setting $\beta = 0$ (no inhibition) yields the same behavior

clumping
gif here

Summary

What is done:

- Traveling waves are solved analytically in the inhibited model.
- Powerful computational framework is built (including GUI)
- Chemotaxis model is introduces and tested
- Within the current framework reproduced: swarming, wave re-initiation, spiral-like waves, CGD-driven clumping, wave decay in 2D-2D setup

Planned:

- Analytical theory of self-extinguishing waves in both 2D-2D and 2D-3D setups
- Analytical theory of clumping (Turing pattern?)
- Biochemical circuit extensions alligned with the experiment
- Generalization to real-tissue geometries